Juan Nicolás Mendoza Roncancio

Engineering Student in Data and Artificial Intelligence at Mines Paris - PSL

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Profile

Msc in Engineering Student at Mines Paris - PSL and Msc in Statiscs, Data Science and IA at Sorbonne Université. Experienced in **Python, PyTorch, SQL and Git** with experience in AI **model deployment and integration**. Recognized for teamwork and adaptability. Seeking a **5-6 month internship starting April 2027** in data and artificial intelligence in France or internationally.

Education

Sorbonne Université **2026 – present**
***Master MS2A (ex M2A)**, Statistics, Data Science and Artificial Intelligence.* *Paris, France*

- Major on **Mathematics and Informatics**.

Mines Paris - PSL **2025 – present**
***Engineering Degree** in Data and Artificial Intelligence* *Paris, France*

- **"Hands on Deep Learning"** course at Télécom Paris (Athens Week exchange)

Universidad Nacional de Colombia **2020 – present**
Bachelor's Degree in Mathematics *Bogotá, Colombia*

- **Top of promotion** · Academic exchange at Universidad de los Andes · **IBM AI Developer** Certificate · Certificate Programs: LLMs, ML, Data Science, HPC/Big Data (2023–24)

Experience

ML Research Engineer – Culstering and Calsification for Medical Images **May 2026 – September 2026**
Center for Applied Mathematics, Mines Paris *Sophia Antipolis, France*

- **Designed a novel medical image classification architecture** combining **clustering image embeddings**, achieving **0.95 accuracy using only 20% of the training data** required by conventional approaches.
- Conducted a **systematic grid search** over training set size and embedding model selection, identifying optimal configurations for data-constrained scenarios.
- Reduced computational cost by **90%** and training time by **100x** by **migrating models to GPU**-accelerated implementations at scale.

ML Engineer – Convex Clustering **March 2025 – May 2026**
Mathematics Department, Universidad de los Andes *Bogotá, Colombia*

- Built a **all-in-one, scikit-learn-ready** clustering library by implementing **seven convex clustering algorithms** behind a single estimator API.
- Ensured reproducible experiment tracking and reliable deployments by setting up **MLflow experiment tracking with S3 artifact storage** and a **CI/CD pipeline with three parallel GitHub Actions jobs**.
- Enabled interactive exploration of clustering results by building a **REST API and Streamlit dashboard** with animated fusion trajectory and weight graph visualization.
- Applied the method to **RFM customer segmentation**, achieving a **clustering quality score of 0.491–0.540** across four data-reduction methods.

ML Engineer - Medical Image Classification **March 2024 – June 2025**
MindLab Group *Bogotá, Colombia*

- Developed **medical image classification** models using **KDM**, achieving an **accuracy of 0.95** comparable to standard architectures.
- Deployed **LLM**-based models such as **MedGemma** on **cloud servers** connected via **Hugging Face** for medical image classification experiments, achieving an **accuracy of 0.98**.

Data Analyst - Energy Efficiency **March 2022 – October 2022**
Universidad Nacional de Colombia *Bogotá, Colombia*

- Identified key energy-efficiency trends by collecting and analyzing data on **500+ high-consumption appliances**.
- Improved dataset reliability by **95%** by designing and executing a data cleaning and preprocessing pipeline.
- Developed decision criteria and presented actionable recommendations to stakeholders, driving a **20% improvement in energy efficiency** and a **15% reduction in energy consumption**.

Projects

Climate & Energy Multi-Agent System | *LangGraph, FastAPI, Redis, GCP* **June 2026**

- Deployed an **8-agent LangGraph pipeline** as a continuous data pipeline on **Google Cloud Run** with real-time messaging, enabling real-time ingestion, quality checks, and issue diagnosis for European energy grid and climate data (ENTSO-E, Copernicus ERA5).
- Ensured system reliability and full monitoring without changing any of the agents' code, by integrating **LangSmith** tracing, an automatic error-recovery system, and an automated testing and deployment pipeline.

Climate & Energy Intelligence — RAG Multi-Agent System | *LangGraph, MongoDB, BERTopic*

- Built a multilingual **AI-powered search pipeline** ingesting 6 real sources by applying a **multilingual topic-clustering technique**, enabling topic detection across different languages without needing a separate model for each language.
- Achieved **79%** agreement on a sentiment classification benchmark by combining **AI-based scoring** with **meaning-based semantic search** and Groq's high-speed AI infrastructure.
- Delivered live climate/energy intelligence to end users via a **4-part AI system (LangGraph)** exposed through a live-updating dashboard, powered by automated nightly data updates.

Skills

Programming Languages: Python: Advanced, **CUDA**: Intermediate, **C++**: Advanced, **Matlab**: Advanced, **Julia**: Intermediate
Frameworks and Libraries: PyTorch: Advanced, **Hugging Face**: Advanced, **Keras**: Advanced, **Scikit-Learn**: Advanced.
Databases: SQL: Intermediate, **MongoDB**: Intermediate, **Cassandra**: Intermediate.
Languages: English: C1, **French**: B2, **Spanish**: Native.